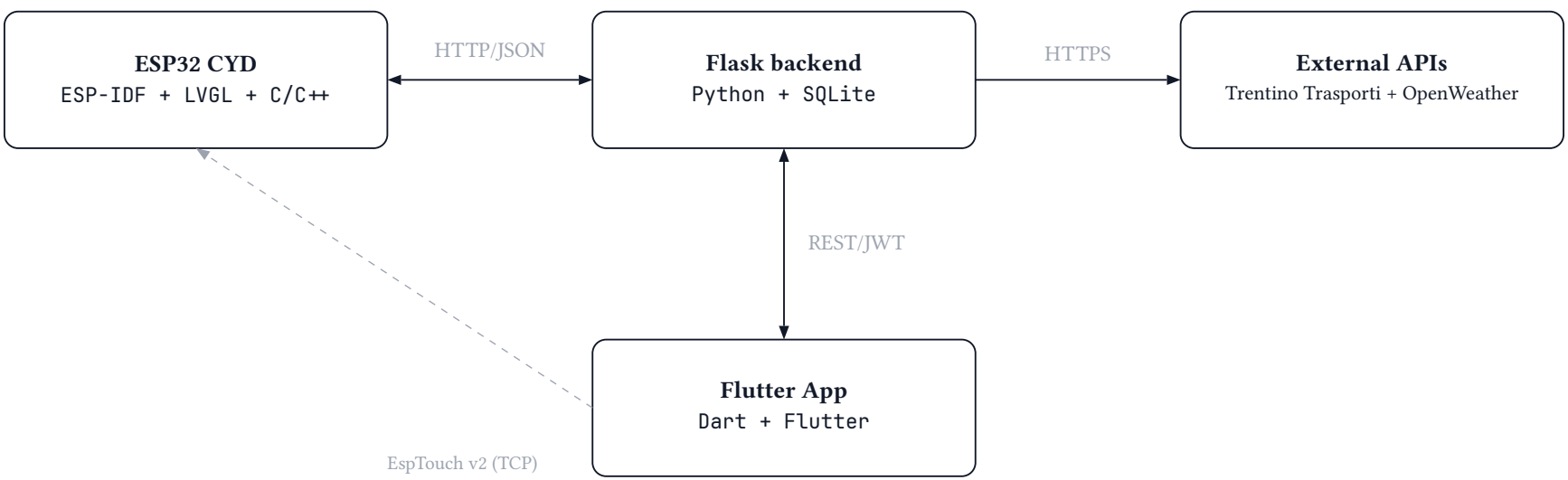
Sdrumo

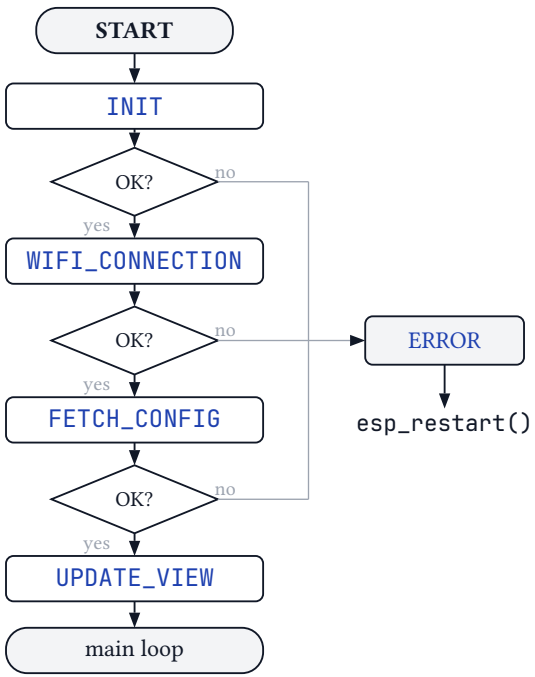
Embedded Software for the Internet of Things

Giulia Cristofolini · Mattia Zagatti · Mirko Lana

Architecture Overview



Software Architecture



C++ singletons

- `Fsm` : state machine
- `Config` : NVS persistence
- `ApiClient` : HTTP client
- `NetHandler` : WiFi + SNTP

FreeRTOS tasks

- `LVGL` port: draw UI on the screen
- `api_poll_task` : bus/weather/config
- `smartconfig_task` : Wi-Fi setup

LVGL screens (C)

- `clock` : analog hands, alarm/timer btns
- `bus` : stop picker + trip list (≤5)
- `weather` : today + 3-day forecast
- `pairing` : token display + done btn

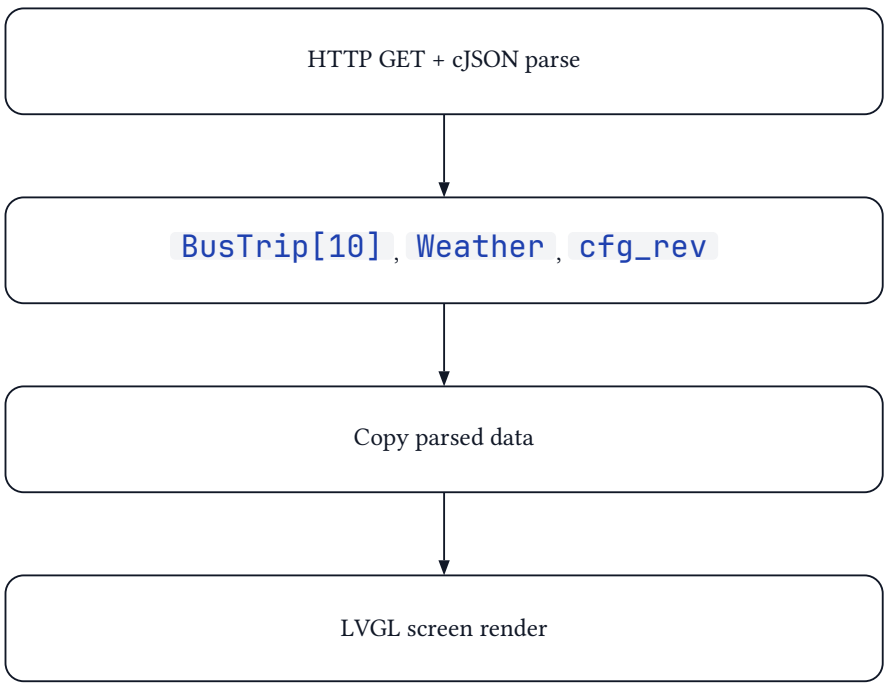
HW/SW Interaction & Data Flow

WiFi events

All hardware events dispatched by ESP-IDF event loop to a single callback. SPI DMA interrupts handled by drivers; app registers only callbacks.

- `STA_START` : connect or SmartConfig
- `STA_DISCONNECTED` : retry or SmartConfig
- `STA_GOT_IP` : `connected = true`
- `SC_GOT_SSID_PSWD` : store credentials, reconnect

Data processing chain



Representative Code

WiFi Event Handler (`net.cpp`)

```
void NetHandler::event_handler(
    void *arg, esp_event_base_t event_base,
    std::int32_t event_id, void *event_data) {
    static NetHandler &net =
NetHandler::get_instance();
    static Config &config  =
Config::get_instance();

    if (event_base == WIFI_EVENT
        && event_id == WIFI_EVENT_STA_START) {
        if (net.smartconfig_running) return;
        if (config.get_ssid()[0U] != '\0'
            && config.get_password()[0U] != '\0')
            esp_wifi_connect();
        else
            net.start_smartconfig();
        return;
    }
    if (event_base == WIFI_EVENT
        && event_id ==
```

LVGL Screen Manager (`screen_manager.c`)

```
void ui_load_screen(lv_obj_t *screen) {
    screen_clock_destroy();
    screen_bus_destroy_timer();
    screen_pairing_destroy_timer();
    lv_obj_clean(screen);

    switch (present_screen_type) {
        case SCREEN_BOOT:
            ui_load_screen_boot(screen); break;
        case SCREEN_WIFI:
            ui_load_screen_wifi(screen); break;
        case SCREEN_CLOCK:
            ui_load_screen_clock(screen);
            ui_load_arrows_btn(screen); break;
        case SCREEN_BUS:
            ui_load_screen_bus(screen);
            ui_load_arrows_btn(screen); break;
        case SCREEN_ALARM:
            ui_load_screen_alarm(screen); break;
        case SCREEN_TIMER:
```

```
WIFI_EVENT_STA_DISCONNECTED) {
        xEventGroupClearBits(net.event_group,
                             NET_CONNECTED_BIT);
        if (net.retry_count < NET_MAX_RETRY)
            { ++net.retry_count;
esp_wifi_connect(); }
        else net.start_smartconfig();
        return;
    }
    if (event_base == IP_EVENT
        && event_id == IP_EVENT_STA_GOT_IP) {
        net.connected = true; net.retry_count =
0;
        xEventGroupSetBits(net.event_group,
                           NET_CONNECTED_BIT);
        if (net.came_from_smartconfig)
            net.came_from_smartconfig = false;
        return;
    }
    if (event_base == SC_EVENT
        && event_id == SC_EVENT_SCAN_DONE)
return;
    if (event_base == SC_EVENT
        && event_id == SC_EVENT_FOUND_CHANNEL)
return;
```

```
            ui_load_screen_timer(screen); break;
        case SCREEN_PAIRING:
            ui_load_screen_pairing(screen);
break;
        case SCREEN_WEATHER:
            ui_load_screen_weather(screen);
            ui_load_arrows_btn(screen); break;
        default:
            ui_load_screen_wifi(screen);
            ui_load_arrows_btn(screen); break;
    }
}
```

```
    if (event_base == SC_EVENT
        && event_id == SC_EVENT_GOT_SSID_PSWD) {
        auto *evt = static_cast<
            smartconfig_event_got_ssid_pswd_t *>(
                event_data);
        char ssid[33U] = { 0 };
        char password[65U] = { 0 };
        memcpy(ssid, evt->ssid, sizeof(evt-
>ssid));
        memcpy(password, evt->password,
               sizeof(evt->password));
        std::size_t sl = strnlen(ssid,
sizeof(ssid)-1);
        std::size_t pl = strnlen(password,
sizeof(password)-1);
        Result cr = config.set_credentials(
            std::string_view(ssid, sl),
            std::string_view(password, pl));
        if (cr != Result::SUCCESS) {
        } else if (config.store() !=
Result::SUCCESS) {}
        if (evt->type == SC_TYPE_ESPTOUCH_V2) {
            uint8_t rvd[RVD_DATA_SIZE] = {};
            ESP_ERROR_CHECK(
                esp_smartconfig_get_rvd_data(
```

```
                    rvd, sizeof(rvd)));
        }
        wifi_config_t cfg = {};
        memcpy(cfg.sta.ssid, evt->ssid,
               sizeof(cfg.sta.ssid));
        memcpy(cfg.sta.password, evt->password,
               sizeof(cfg.sta.password));
        net.retry_count = 0;
        ESP_ERROR_CHECK(
            esp_wifi_set_config(WIFI_IF_STA,
&cfg));
        ESP_ERROR_CHECK(esp_wifi_connect());
        return;
    }
    if (event_base == SC_EVENT
        && event_id == SC_EVENT_SEND_ACK_DONE) {
        xEventGroupSetBits(net.event_group,
NET_ESPTOUCH_DONE_BIT);
        return;
    }
}
```

Testing & Conclusions

Testing

- `idf.py monitor`: FSM states, HTTP payloads, task logs
- Bruno collection: testing all APIs before their implementation in the firmware
- Physical device: checking if actual result = desired result
- `flutter analyze` + Android/Linux run
- `idf.py erase-flash`: full re-registration and re-pairing + Wi-Fi configuration

Result

Working embedded product: live busses + weather on a 2.8″ touch display; zero-config WiFi pairing.

Future improvements

- HTTPS support
- MQTT / WebSocket to replace config polling
- Light-sleep + screen off during night
- Alarm & timer backend (screens already done)
- Buzzer for timer and alarm